Ethan Rong

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Work Experience

Wealthsimple Jan. 2027 – Apr. 2027
Software Engineer Intern, Platform Engineering *Toronto, ON*

- Planning on doing a 16 week internship on the platform engineering team, focusing on scaling infrastructure.

Meta Jun. 2026 – Aug. 2026
Software Engineer Intern, AI Infrastructure *Seattle, WA*

- Designed a serverless service for creating and deploying pre-runtime policies that **cut 12K+ hrs of wasted GPU wall time** by preventing crash-prone jobs across **1.5M+ requests/day** from claiming GPU capacity.
- Saved **$13K+ in GPU compute** with an AI agent proposing 6 policies from failed jobs, gated by 12 checks.
- Built an Airflow (Dataswarm) ETL pipeline aggregating GPU job failures **across 100K daily jobs and 79 policies** into Hive-backed datasets and reporting tooling; authored 4 design docs defining the pipeline architecture.
- **As primary on-call** for the agentic policy service, which pushes code to prod daily, defined **6 SLIs** and alerts in Prometheus/Grafana (ODS), and added a kill switch, ACL guards, and an idempotency guard.
- Diagnosed out-of-memory crashes in a Python service on Kubernetes (Tupperware) to compile-time spikes and scaled container RAM from **32GB to 48GB**, cutting peak memory pressure from **90% to 70% (1.3x)**.
- Improved policy validation latency **from 12s to 3.5s (3.4x)** by parallelizing a validation pipeline.

Meta May 2025 – Aug. 2025
Software Engineer Intern, Reliability Infrastructure *Menlo Park, CA*

- Shipped RPM-packaged CLI commands for a fault-injection service handling **790K+ requests/year across 85 services**, that automates metadata resolution and workflows with ∼**290 tests at 90% unit test coverage**.
- Deployed **React/TypeScript** fault-injection UI to **136 engineers in 15 teams**, enabling **95%+ of use cases**.
- Engineered safe network block faults for **thousands of Tupperware jobs** via Python task-to-host IP mapping.
- Rewrote a fault injection CLI in Rust, deployed across a **Kubernetes (Tupperware) fleet of 100K+ dev machines**; authored design docs and demoed to senior leadership.

Western University Feb. 2025 – May 2025
Research Assistant, Ivey Business School *London, ON*

- Published a case study on DeepSeek's open-source AI strategy, **taught to 1,000+ students in strategy, tech, and finance courses**; evaluated 3 monetization models for the financial feasibility of open-source models.
- Guided AI vendor choice by benchmarking DeepSeek vs. ChatGPT on 3 use cases, finding better token efficiency.

Major League Hacking (MLH) Jun. 2024 – Sep. 2024
Site Reliability Engineer (SRE) Fellow, S24 *New York, NY*

- Accelerated deployment times by using GitHub Actions and Bash, and optimized a Python-Flask web app by shrinking Docker container size from 2.8GB to 100MB through layer caching, enabling zero-downtime deployments.

Projects

CityFlow | *Python, NVIDIA Nemotron, DGX Spark, LangChain, FastAPI* NVIDIA Spark Hackathon

- Won **1st out of 100+ teams** in the Economic System track for an on-device AI agent that generates actionable insights for small businesses; **live-streamed demo to 700+ viewers during an NVIDIA livestream event.**

Hack Western 12 Badge System | *Next.js, TypeScript, tRPC, PostgreSQL* Hack Western 12 Web Org.

- Launched **Hack Western's badge system** serving 500+ attendees, replacing 600+ badges with Apple Wallet.

Technical Skills

Languages/Frameworks: Python, TypeScript, Rust, Java, C, C++, Express.js, Flask, Node.js, Nginx, Claude Code
Infra: Kubernetes, Prometheus/Grafana, Hive, AWS (S3, Lambda, VPC), GCP (Cloud Run/Function), Linux, Docker
Databases: PostgreSQL, MySQL, SQLite, MongoDB, Redis

Education

University of Western Ontario London, ON
BSc Computer Science & Honors Business Administration, GPA: 3.7/4.0 *Expected Dec. 2027*

- **Coursework:** Operating Systems (Linux), Networks, Distributed Systems, Databases, Data Struct. & Algo.
- **Clubs:** Hack Western (Web Dev), Western Founders' Network (VP), Western Tech Alumni (Organizer)